

HR Analytics Report

Summary

This report analyzes workforce data from a company with 311 employees across multiple departments and positions. The analysis reveals key insights about employee demographics, compensation patterns, and organizational structure that can inform strategic HR decisions.

Variable Definitions:

The dataset includes comprehensive employee information covering personal identifiers (Employee_Name, EmpID), demographic details (GenderID/Sex, MarriedID/MaritalStatusID, DOB, CitizenDesc, HispanicLatino/RaceDesc), and geographic data (State/Zip). Employment-related variables encompass current status indicators (EmpStatusID/EmploymentStatus, Termd), organizational structure (DeptID/Department, PositionID/Position), management relationships (ManagerName/ManagerID), and recruitment information (RecruitmentSource, FromDiversityJobFairID).

Performance and engagement metrics include evaluation scores (PerfScoreID/PerformanceScore), survey results (EngagementSurvey, EmpSatisfaction), project involvement (SpecialProjectsCount), and review timing (LastPerformanceReview_Date). Compensation data is captured through annual Salary figures, while behavioral indicators track attendance patterns (DaysLateLast30, Absences). Employment timeline information includes hiring and departure dates (DateofHire/DateofTermination) along with termination reasons (TermReason).

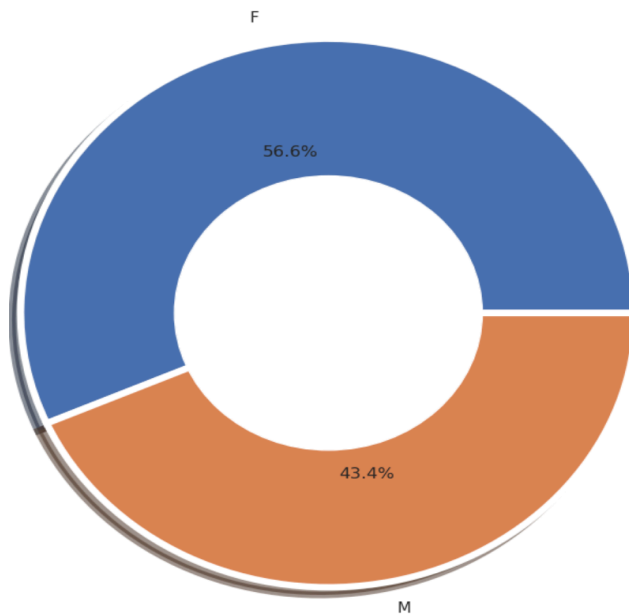
Key Findings

Workforce Composition

- **Gender Distribution:** The workforce is fairly balanced with a slight female majority (56.6% female, 43.4% male)

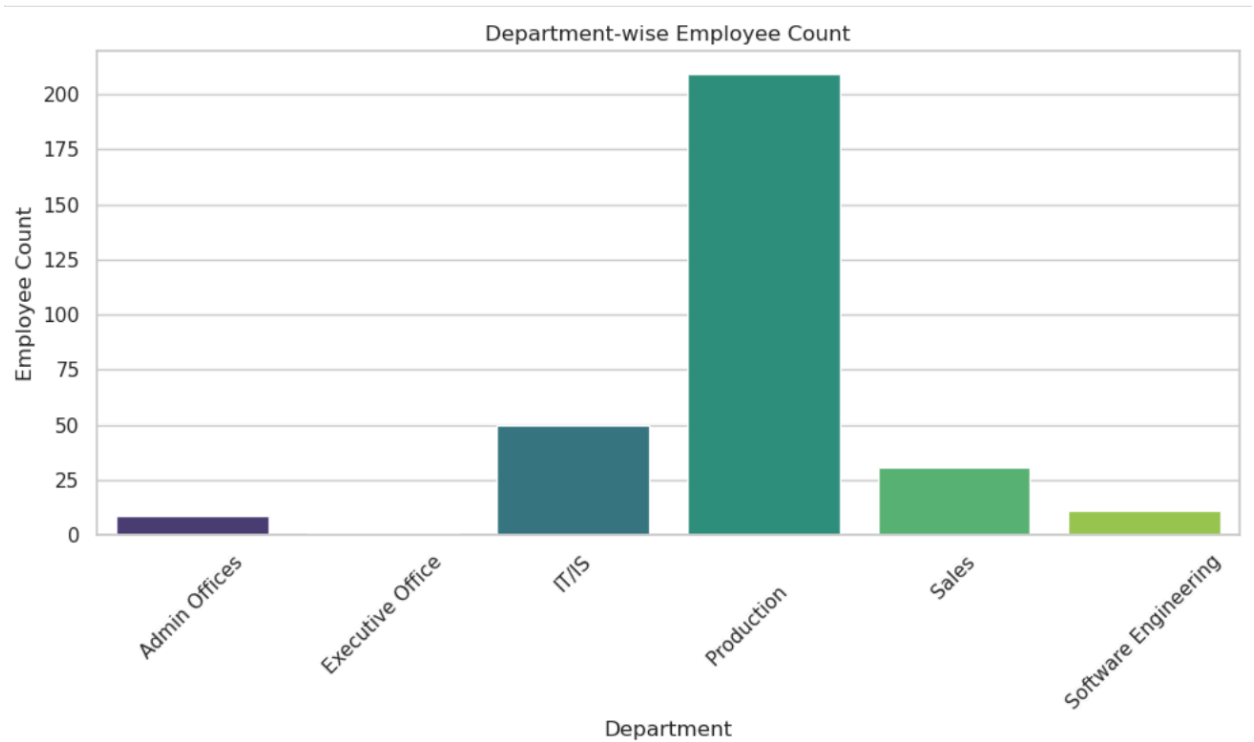
Gender Distribution Pie Chart

Female vs Male Distribution



- **Department Concentration:** Production dominates with over 200 employees, followed by IT/IS (~50 employees), while other departments (Admin, Executive, Sales, Software Engineering) have smaller teams

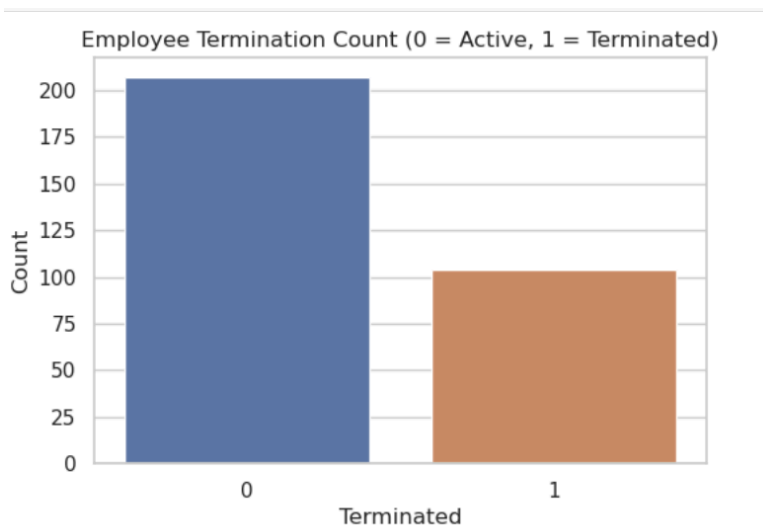
Department-wise Employee Count Bar Chart



Employee Retention

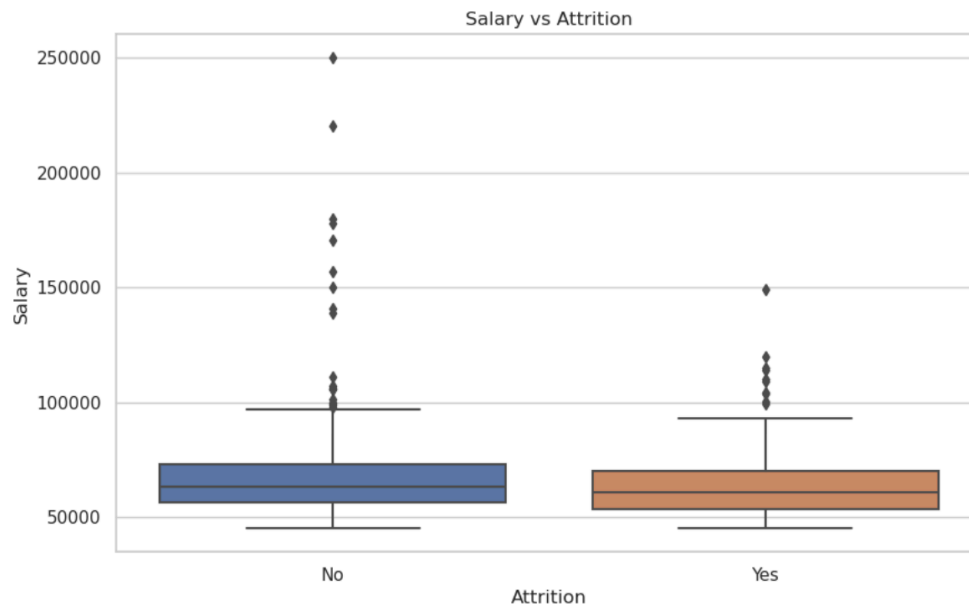
- **Retention Rate:** Approximately 67% of employees remain active, with 33% having been terminated

Employee Termination Count Chart



- **Attrition Patterns:** The salary distribution between current and former employees shows similar ranges, suggesting compensation isn't the primary driver of turnover

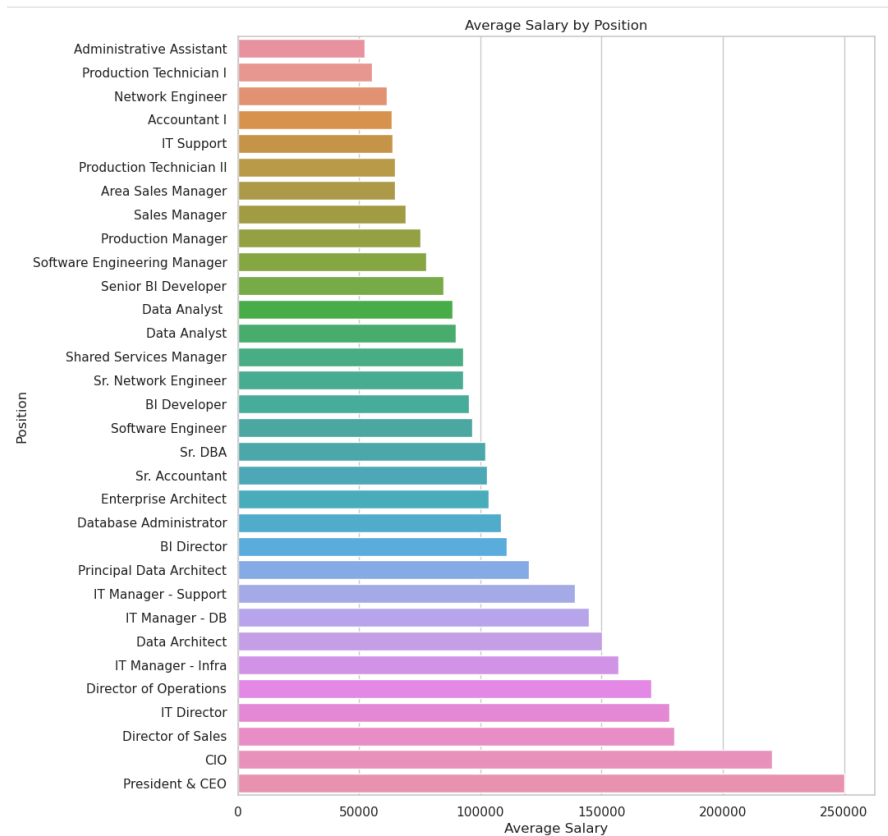
Salary vs Attrition Box Plot



Compensation Structure

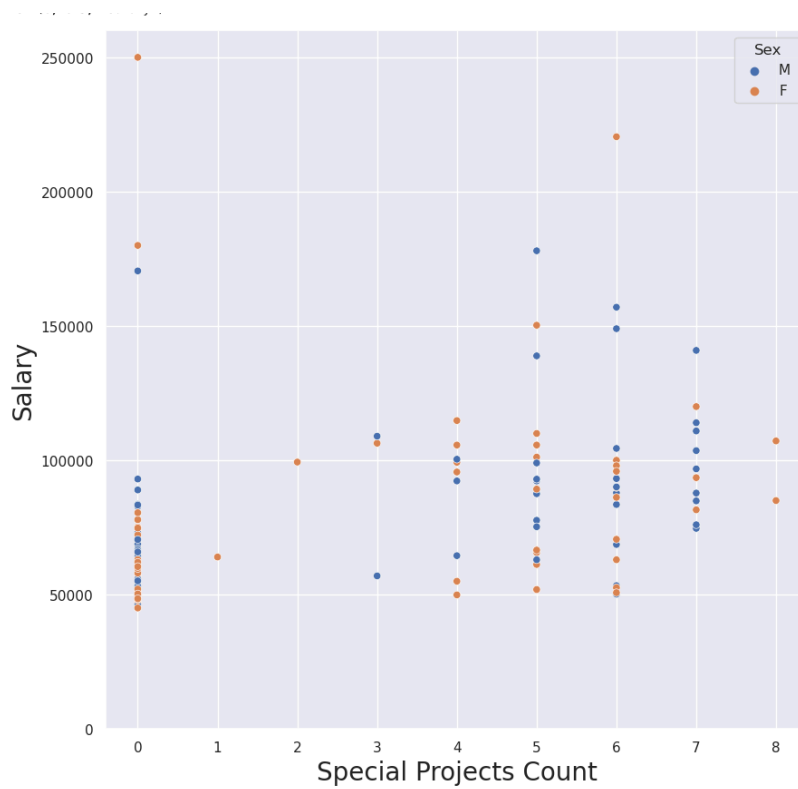
- **Salary Range:** Positions show significant variation, from entry-level roles around \$45K to executive positions reaching \$250K+
- **Leadership Premium:** C-suite positions (President & CEO, CIO) command the highest salaries, followed by IT Directors and Director-level roles

Average Salary by Position Bar Chart



- **Special Projects Impact:** There's a positive correlation between involvement in special projects and salary levels, particularly noticeable among higher-paid employees

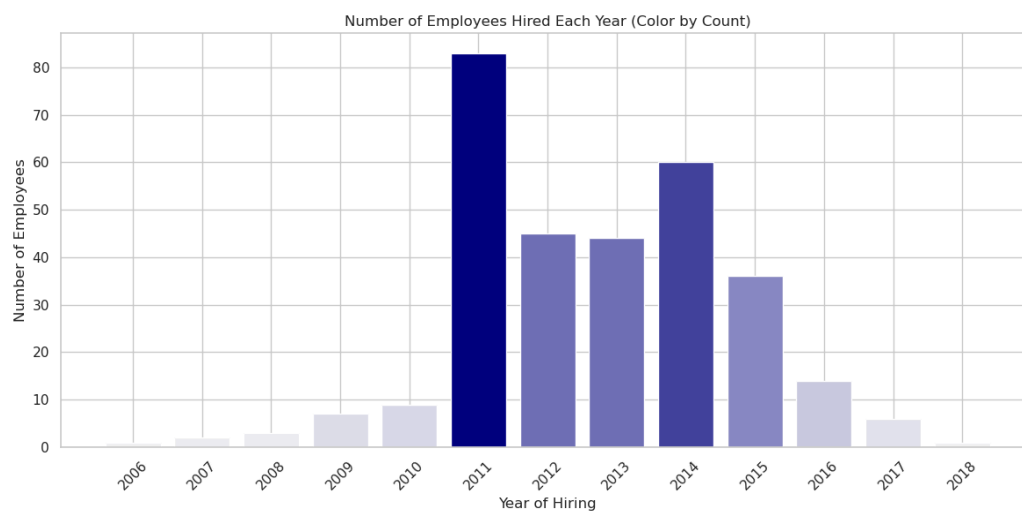
Special Projects Count vs Salary Scatter Plot



Hiring Trends

- **Peak Hiring:** 2011 saw the highest recruitment activity with over 80 new hires
- **Recent Patterns:** Hiring has generally declined since 2014, with minimal recruitment in recent years (2017-2018)

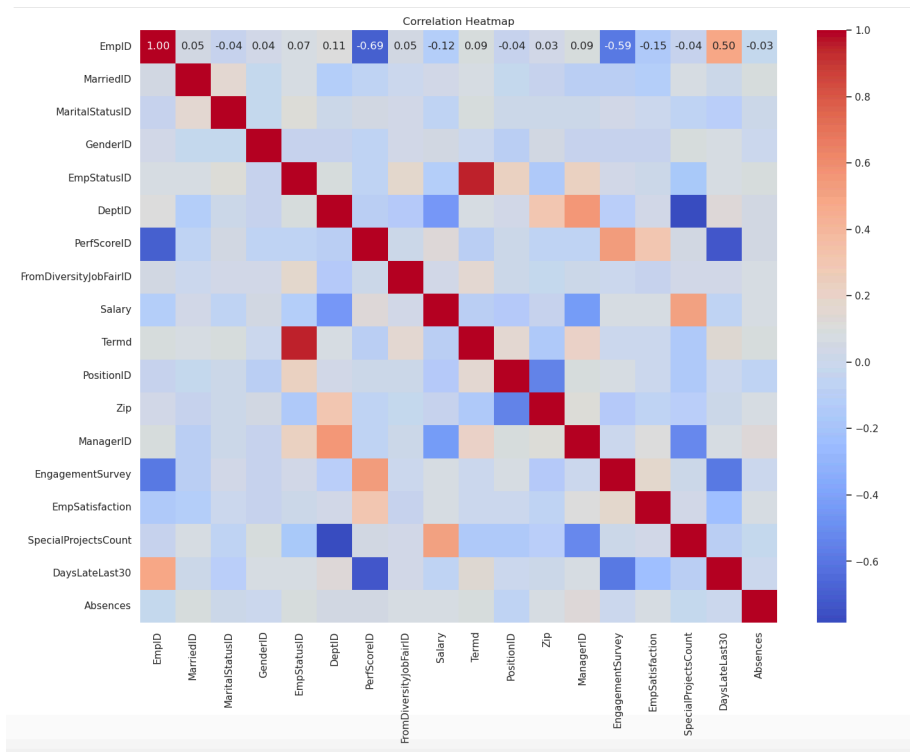
Number of Employees Hired Each Year



Data Relationships

The correlation analysis reveals interesting patterns across various employee metrics, showing how different factors interact within the organization.

Correlation Heatmap



Salary Correlations: Compensation shows strong positive correlations with position level and department type, which aligns with the expected organizational hierarchy

Performance Clustering: Employee satisfaction, engagement survey scores, and special project involvement cluster together with positive correlations, indicating that highly engaged employees tend to take on greater responsibilities

Attendance Patterns: Days late and absence counts show positive correlation, reflecting consistent patterns in employee attendance behavior

Turnover Risk Indicators: Termination status negatively correlates with several performance and satisfaction metrics, suggesting these could serve as early warning signs for potential departures

Demographic Neutrality: Gender, marital status, and other demographic variables show minimal correlation with performance indicators, demonstrating relatively fair and unbiased organizational practices

Strategic Recommendations

1. **Retention Focus:** With a 33% termination rate, investigate exit interview data to identify improvement opportunities in employee satisfaction and engagement
2. **Departmental Balance:** Consider cross-training initiatives to reduce over-reliance on the Production department and build organizational resilience
3. **Career Development:** Leverage the positive correlation between special projects and compensation to create clearer advancement pathways
4. **Recruitment Strategy:** Evaluate whether recent low hiring levels align with business growth objectives, particularly given the aging workforce from the 2011-2014 hiring surge

Conclusion

The organization shows a healthy gender balance and clear compensation structure. However, attention should be paid to retention rates and ensuring adequate talent pipeline through strategic hiring. The strong correlation between project involvement and compensation suggests a merit-based culture that can be further leveraged for employee development and retention.

Reference

Human Resources Data set. (2020, October 19). Kaggle.

<https://www.kaggle.com/datasets/rhuebner/human-resources-data-set>